LLMs & VLMs course

The attention mechanism

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Contents

Motivation

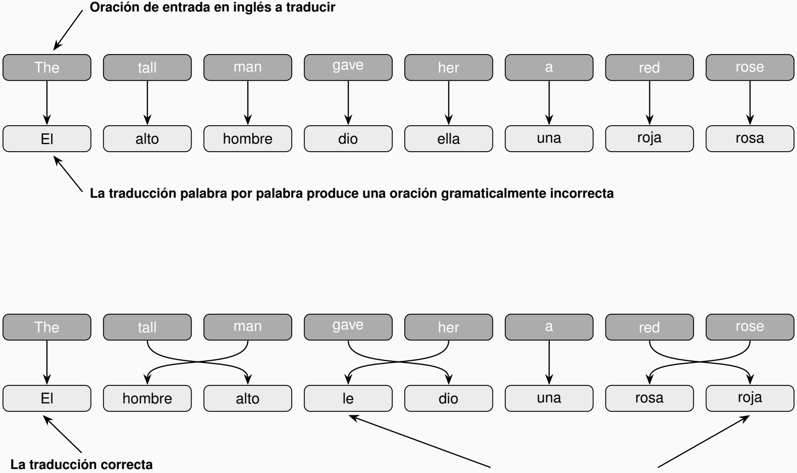
The attention mechanism

Roadmap

Motivation

The attention mechanism

The Motivation for Attention



Oración de entrada en inglés a traducir

The	tall	man	gave	her	a	red	rose
El	alto	hombre	dio	ella	una	roja	rosa

La traducción palabra por palabra produce una oración gramaticalmente incorrecta

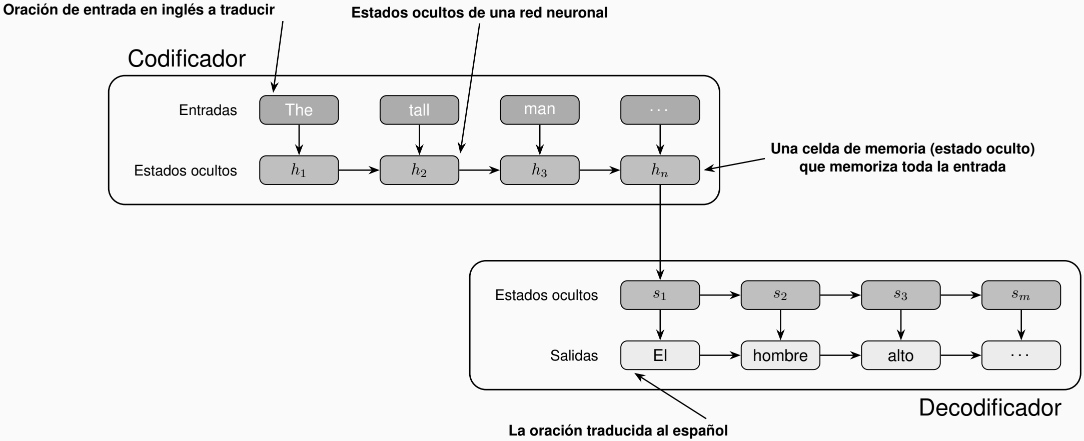
The	tall	man	gave	her	a	red	rose
El	hombre	alto	le	dio	una	rosa	roja

La traducción correcta

Ciertas palabras de la traducción generada requieren acceso a palabras que aparecen antes o después en la oración original

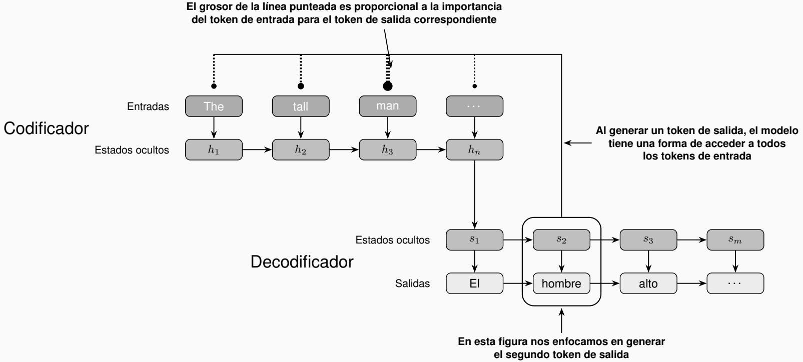
The Motivation for Attention

- **Problem with RNNs:** Difficulty in capturing long-range


Oración de entrada en inglés a traducir
Estados ocultos de una red neuronal
Codificador
Entradas
The
tall
man
· · ·
Estados ocultos
h_1
h_2
h_3
h_n
Una celda de memoria (estado oculto) que memoriza toda la entrada
Estados ocultos
s_1
s_2
s_3
s_m
Salidas
El
hombre
alto
· · ·
Decodificador
La oración traducida al español

The Motivation for Attention

- **The Attention Concept:** Allows the model to focus on different parts of the input sequence regardless of their distance.
- **Key Insight:** Instead of a fixed-size hidden state, we compute a weighted sum of all input tokens' representations.



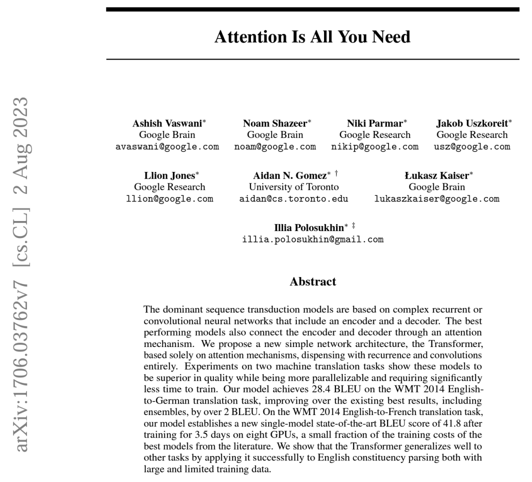
El grosor de la línea punteada es proporcional a la importancia del token de entrada para el token de salida correspondiente

Al generar un token de salida, el modelo tiene una forma de acceder a todos los tokens de entrada

En esta figura nos enfocamos en generar el segundo token de salida

Entradas
The
tall
man
· · ·
Codificador
Estados ocultos
h_1
h_2
h_3
h_n
Estados ocultos
s_1
s_2
s_3
s_m
Decodificador
Salidas
El
hombre
alto
· · ·

The paper that made the revolution



arXiv:1706.03762v7 [cs.CL] 2 Aug 2023

Attention Is All You Need

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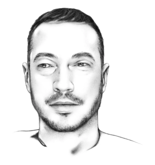
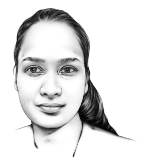
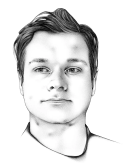
Abstract

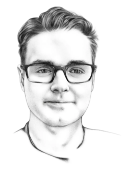
The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

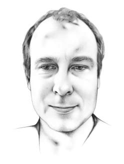
8 Google Employees Invented Modern AI[1]

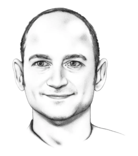

(a) J. Uszkoreit

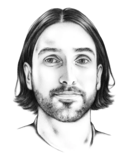

(b) Niki Parmar

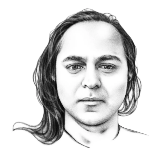

(c) I. Polosukhin.


(d) Llion Jones.


(e) L. Kaizer.


(f) N. Zhazeer.


(g) A.N. Gomez.


(h) A. Vaswani.

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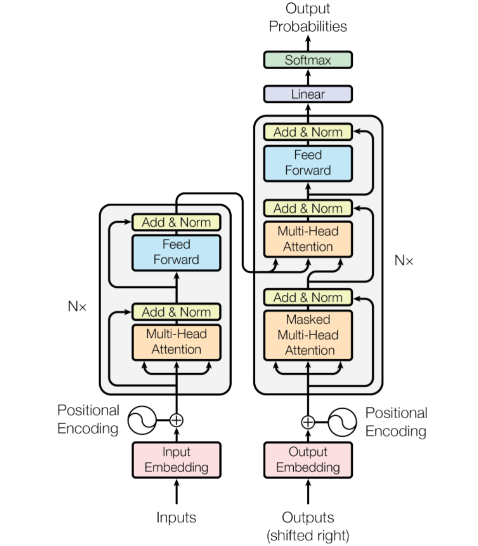
[1] Steven Levy, "8 Google Employees Invented Modern AI. Here's the Inside Story," Wired, 20 de marzo de 2024

The transformer architecture


Output
Probabilities
Softmax
Linear
Add & Norm
Feed
Forward
Add & Norm
Multi-Head
Attention
Nx
Add & Norm
Masked
Multi-Head
Attention
Add & Norm
Feed
Forward
Nx
Add & Norm
Multi-Head
Attention
Positional
Encoding
Positional
Encoding
Input
Embedding
Output
Embedding
Inputs
Outputs
(shifted right)

Roadmap

Motivation

The attention mechanism

Initial explanation

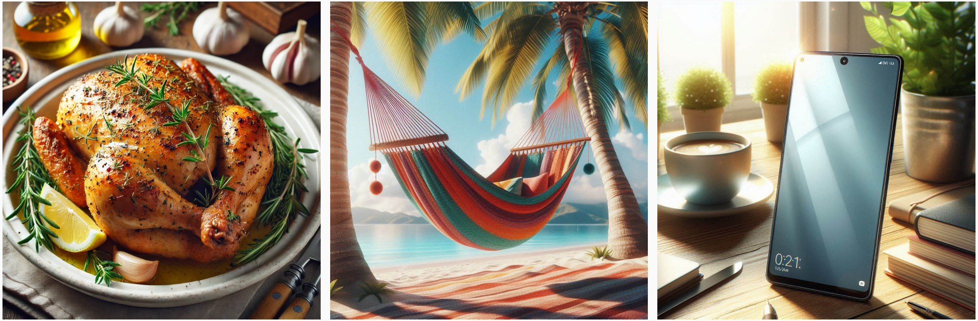


Figure 2: Volitional vs non-volitional cues

From Input to Attention Spaces

The input matrix X (comprising token embeddings and positional encodings) is linearly projected into three separate vector spaces:

Query (Q)

What am I looking for?

Represents the current token's focus or "question" directed at the rest of the sequence.

Key (K)

What do I offer?

Represents the relevance label of a token. It is used to match against incoming Queries.

Value (V)

What do I contain?

Holds the actual semantic content that will be extracted if a strong match occurs.

Mathematical Formulation

Each matrix is generated by multiplying the input matrix X by its respective learnable weight matrix:

$$Q = X \cdot W^Q \tag{1}$$

$$K = X \cdot W^K \tag{2}$$

$$V = X \cdot W^V \tag{3}$$

The projection weight matrices W^Q, W^K, and W^V are randomly initialized and optimized end-to-end via backpropagation during training.

Attention mechanism (Vaswani et al. 2017)

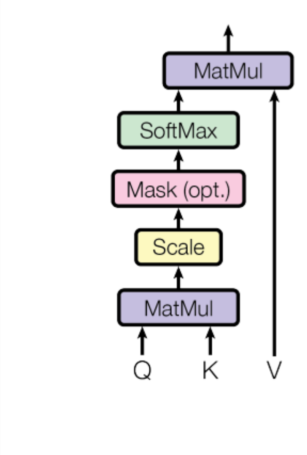

MatMul
SoftMax
Mask (opt.)
Scale
MatMul
Q
K
V

Figure 3: Scaled dot-product attention

Scaled Dot-Product Attention: The Math

The core operation involves three learnable matrices: **Query (Q)**, **Key (K)**, and **Value (V)**.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{4}$$

- **Dot-Product (QK^T):** Measures similarity between queries and keys.
- **Scaling ($1/\sqrt{d_k}$):** Prevents the softmax from entering regions with extremely small gradients when d_k is large.
- **Softmax:** Normalizes scores into probabilities (weights summing to 1).

Multi-Head Attention (MHA)

Why use multiple heads?

- Allows the model to jointly attend to information from different representation subspaces at different positions.
- **Process:**
  1. Split Q, K, V into h heads.
  2. Perform attention in parallel.
  3. Concatenate the results and apply a linear projection.

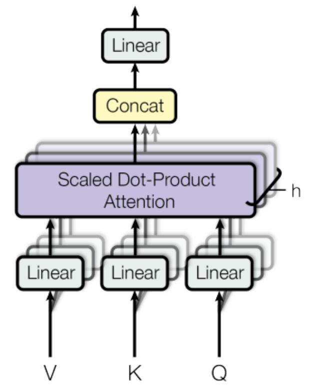

Linear
Concat
Scaled Dot-Product Attention
h
Linear
Linear
Linear
V
K
Q

Figure 4: Multi-Head Attention consists of several attention layers running in parallel

Multi-head attention

$$\mathrm{MultiHeadAttention}(Q, K, V) = \mathrm{Concat}(\mathrm{head}_1, \mathrm{head}_2, \ldots, \mathrm{head}_h)W^O \tag{5}$$

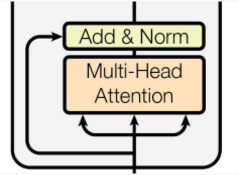
where

$$\mathrm{head}_i = \mathrm{Attention}(QW_i^Q, KW_i^K, VW_i^V) \tag{6}$$

Self-Attention vs. Cross-Attention

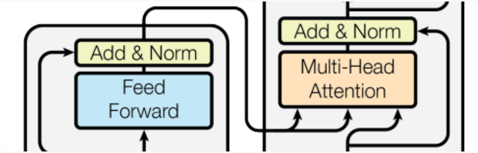
Self-Attention:

- Q, K, V all originate from the same input sequence.
- Used in Encoders and Decoders to relate different positions of the same sequence.


Add & Norm
Multi-Head Attention

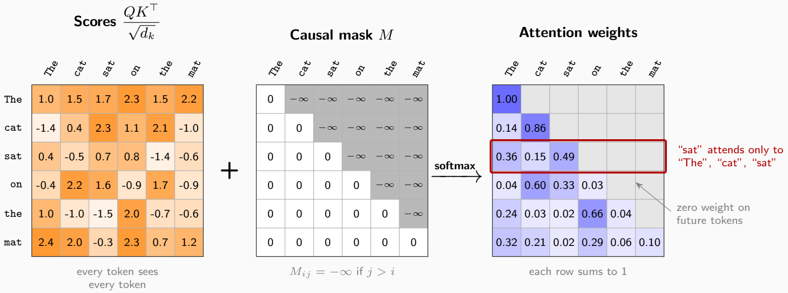
Cross-Attention:

- Q comes from one sequence (e.g., decoder), while K and V come from another (e.g., encoder output).
- Essential for tasks like Machine Translation.


Add & Norm
Feed Forward
Add & Norm
Multi-Head Attention

Causal Attention: Masking the Future

In GPT-like models (Decoder-only), a token should only attend to previous tokens.



Scores $\frac{QK^\top}{\sqrt{d_k}}$

	The	cat	sat	on	the	mat
The	1.0	1.5	1.7	2.3	1.5	2.2
cat	-1.4	0.4	2.3	1.1	2.1	-1.0
sat	0.4	-0.5	0.7	0.8	-1.4	-0.6
on	-0.4	2.2	1.6	-0.9	1.7	-0.9
the	1.0	-1.0	-1.5	2.0	-0.7	-0.6
mat	2.4	2.0	-0.3	2.3	0.7	1.2

every token sees every token

$+$

Causal mask M

The	cat	sat	on	the	mat
0	$-\infty$	$-\infty$	$-\infty$	$-\infty$	$-\infty$
0	0	$-\infty$	$-\infty$	$-\infty$	$-\infty$
0	0	0	$-\infty$	$-\infty$	$-\infty$
0	0	0	0	$-\infty$	$-\infty$
0	0	0	0	0	$-\infty$
0	0	0	0	0	0

$M_{ij} = -\infty$ if $j > i$

Attention weights

The	cat	sat	on	the	mat
1.00					
0.14	0.86				
0.36	0.15	0.49			
0.04	0.60	0.33	0.03		
0.24	0.03	0.02	0.66	0.04	
0.32	0.21	0.02	0.29	0.06	0.10

“sat” attends only to “The”, “cat”, “sat”

zero weight on future tokens

each row sums to 1

$$\text{CausalAttention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + M\right)V$$

When is it used?

✓ **Decoder-only LLMs** (GPT-2, GPT-3, Llama): every self-attention layer.
✓ **Decoder of the original Transformer**: masked self-attention.
✓ **Training:** predict all next tokens in parallel without “peeking” at the answer.
✓ **Generation:** each new token depends only on the previous ones.

When is it not used?

× **Encoders** (BERT, ViT): bidirectional, every token sees all tokens.
× **Cross-attention** (decoder → encoder output): the whole source is already known.

The <|pad|> mask is a different mask: it hides padding positions, not the future.

Computational Efficiency and Complexity

- **Self-Attention Complexity:** $O(L^2 \cdot d)$, where L is sequence length and d is embedding dimension.
- **Comparison:**
  - RNNs are $O(L \cdot d^2)$ but sequential.
  - Transformers are parallelizable but quadratic with respect to length.
- This quadratic growth is why extending the **Context Window** is a major research challenge in LLMs.

Practice: Implementing the attenttion mechanism